$= [a_1 + a_2 \ldots\ldots an] + ([b_1+c_1 \;\; b_2+c_2 \ldots\ldots b_n+c_n])$

$= [a_1 \; a_2 \ldots\ldots a_n] + ([b_1 \; b_2 \ldots\ldots b_n] + [c_1 \; c_2 \ldots\ldots c_n])$

$= A+(B+C)$

c) $A+0 = [a_1+0 \;\; a_2+0 \ldots\ldots a_n+0] = [a_1 \; a_2 \ldots\ldots a_n]$ (Algebraic property 3)

$= A$

d) $r(A+B) = r[(a_1+b_1) \; (a_2+b_2) \ldots\ldots (a_n+b_n)] = [ra_1+rb_1 \;\; ra_2+rb_2 \ldots\ldots ra_n+rb_n]$ (Algebraic property v) $= [ra_1 \; ra_2 \ldots\ldots ra_n] + [rb_1 \; rb_2 \ldots\ldots rb_n]$

$= r[a_1 \; a_2 \ldots\ldots a_n] + r[b_1 \; b_2 \ldots\ldots b_n] = rA+rB$

e) $(r+s)A = (r+s)[a_1 \; a_2 \ldots\ldots a_n] = [ra_1+sa_1 \;\; ra_2+sa_2 \ldots\ldots ra_n+sa_n]$ (Algebraic property vi) $= [ra_1 \; ra_2 \ldots\ldots ra_n] + [sa_1 \; sa_2 \ldots\ldots sa_n]$

$= rA+sA$

f) $r(sA) = r(s[a_1 \; a_2 \ldots\ldots a_n]) = r[sa_1 \; sa_2 \ldots\ldots sa_n] = [r(sa_1) \; r(sa_2) \ldots\ldots r(sa_n)] = [(rs)a_1 \; (rs)a_2 \ldots\ldots (rs)a_n]$ (Algebraic property vii)

$= (rs)A$ □

Matrix Multiplication

When a matrix B multiplies a vector x, it transforms x into the vector Bx. If this vector is then multiplied in turn by a matrix A, the resulting vector is $A(Bx)$.

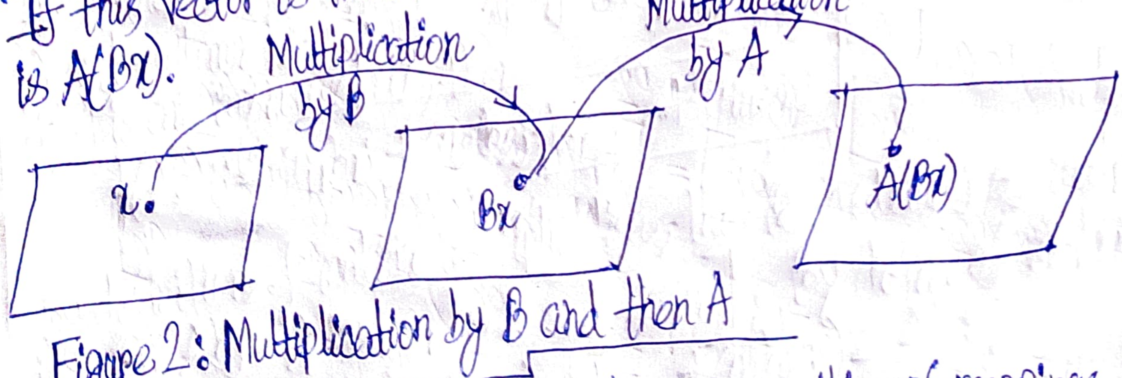

Figure 2: Multiplication by B and then A

Thus $A(Bx)$ is produced from x by a composition of mappings. Our goal is to represent this composite mapping as multiplication by a single matrix, denoted by AB, so that $\boxed{A(Bx) = (AB)x \quad (1)}$